

Google Step Up Challenge



Your Mission: Power Google's Gemini Pro Breakthrough

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What trends exist in the campaign data?

Which markets and channels performed best? (ie. had the most sales and best efficiency?)

The **SA** market with **Youtube** channel campaign is the most efficient as it achieved and CPLU (Cost Per Lifted User) of \$0.45, which is **13%** better than the next best market EG with Social channel.

--- Performance Summary by Market & Channel ---					
	Market	Channel	Spend_USD	Lifted_users	CPLU
11	SA	YouTube	586347	1.298375e+06	0.451601
6	EG	Social	599752	1.182176e+06	0.507329
7	EG	YouTube	579792	1.136869e+06	0.509990
3	DE	YouTube	885143	1.664343e+06	0.531827
10	SA	Social	577913	1.054470e+06	0.548060
2	DE	Social	853339	1.458862e+06	0.584935
14	UK	Social	892537	1.512776e+06	0.590000
15	UK	YouTube	911210	1.499302e+06	0.607756
4	EG	Display	390429	3.644865e+05	1.071175
8	SA	Display	335932	2.834696e+05	1.185072
12	UK	Display	548186	1.520505e+05	3.605289
0	DE	Display	546337	1.511458e+05	3.614635
5	EG	Search	581230	0.000000e+00	inf
1	DE	Search	903083	0.000000e+00	inf
9	SA	Search	583448	0.000000e+00	inf
13	UK	Search	842667	0.000000e+00	inf

What is the relative cost to target by market/channel?

The **UK** market with **search** channel is the most expensive to target. However, if we look at the overall data, notice that the Search channel CPR (Cost per Reach) are about **\$0.09** and the CPR for Display channel are about \$0.019 regardless of where the market is. This means that the Search channel cost more and that the **channel is more important** to consider than the market.

--- Relative Cost to Target by Market/Channel ---				
	Market	Channel	CPR	Relative_Cost_Index
13	UK	Search	0.093284	203.252362
9	SA	Search	0.092691	201.960476
5	EG	Search	0.092428	201.387878
1	DE	Search	0.092347	201.211206
2	DE	Social	0.036137	78.738294
10	SA	Social	0.036045	78.537291
15	UK	YouTube	0.036006	78.451298
11	SA	YouTube	0.035922	78.269933
6	EG	Social	0.035826	78.060717
14	UK	Social	0.035251	76.806551
3	DE	YouTube	0.034920	76.085745
7	EG	YouTube	0.034717	75.643653
4	EG	Display	0.019769	43.073415
0	DE	Display	0.019731	42.991981
12	UK	Display	0.019721	42.968412
8	SA	Display	0.019533	42.560787

Which things worked best for brand lift?

What campaigns were most effective (drove the most relative lift)

The most effective and statistically significant campaign is BackToSchool_24 in the SA market with the Youtube Channel, with a relative lift of 103.93 which is the highest. We can also concur from the result that BackToSchool_24 (2024) is better than BackToSchool_23 (2023) with the Display channel, which have the lowest relative lift, this means that the campaign improved over time and could also mean that the Youtube Channel is much more effective in the SA market.

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--- Most Effective and Statistically Significant campaigns (Highest Relative Lift) ---
  Campaign_Name Market Channel Relative_Lift
1432 BackToSchool_24    SA  YouTube    103.931185
1416 BackToSchool_24    SA  YouTube    103.931185
1400 BackToSchool_24    SA  YouTube    103.931185
1368 BackToSchool_24    SA  YouTube    103.931185
1384 BackToSchool_24    SA  YouTube    103.931185
...      ...      ...      ...
634  BackToSchool_23    SA  Display     18.427706
538  BackToSchool_23    SA  Display     18.427706
618  BackToSchool_23    SA  Display     18.427706
522  BackToSchool_23    SA  Display     18.427706
602  BackToSchool_23    SA  Display     18.427706
```

Which were statistically significant?

Bonus: Which was most efficient?

Determined by cost per lifted user (CPLU), the lesser the cost means the more efficient. ExamPrep_23 with the Youtube Channel in the SA market have a 0.15 CPLU which is the lowest, which means it is the most efficient.

```
Top 5 Most Efficient Campaigns:
  Campaign_Name Market Channel cost_per_lifted_user
280    ExamPrep_23    SA  YouTube      0.155394
620  BackToSchool_23    EG  YouTube      0.168491
537  BackToSchool_23    SA   Social      0.172854
556  BackToSchool_23    EG  YouTube      0.177567
1380 BackToSchool_24    DE  YouTube      0.185194
```

What creatives (adverts) resonated best with our target audience?

The creative that resonates best is **Life Hack** with a global average lift of **5.51** which is **5x more effective** than Free trial which have 1.06 global average lift.

```
--- Global Creative Resonance (Avg Consideration Lift) ---
Creative_Name
Life Hack      5.512917
Coding Help    3.079583
Study With Me  2.261875
Grade Booster  1.626250
Free Trial      1.065000
```

Looking at the performance in each market of each creative. We can see that the Life Hack creative resonate very well in the **DE** and **UK market** with a **6.1-6.3** point est. consideration, but doesn't perform as well in the EG and SA market. Additionally, **Coding Help** perform quite well in the **EG** and **SA market** with a **3.51 and 3.37 respectively**. We could use Life Hack in DE and UK market and Coding Help in EG and SA market.

```
--- Resonance by Market ---
Creative_Name  Coding Help  Free Trial  Grade Booster  Life Hack  Study With Me
Market
DE             2.722500    1.078333    1.660000    6.313333    2.210000
EG             3.512500    1.072500    1.583333    4.724167    2.320833
SA             3.370000    1.058333    1.624167    4.895833    2.285833
UK             2.713333    1.050833    1.637500    6.118333    2.230833
```

The channel that resonates best is **Youtube** with a **4.53** point est. consideration, followed by **Social** with a **4.49** point est. consideration.

```
--- Resonance by Channel ---
Channel
YouTube      4.536333
Social       4.499667
Display      1.191833
Search       0.608667
```